CSE 331 - Quiz 3

Q1. Let $s = 0^p 1 0^p 1 0^p 1$

$\underbrace{0^\alpha}_{x} \underbrace{0^\beta}_{y} \underbrace{0^{p-\alpha-\beta} 1 0^p 1 0^p 1}_{z}$

pumping i times,

we get, $0^\alpha 0^{i\beta} 0^{p-\alpha-\beta} 1 0^p 1 0^p 1$

$= 0^{p+i\beta-\beta} 1 \, 0^p 1 \, 0^p 1$

The three equal parts must contain one 1. Thus, to prove non-regularity we need to show the 0's are not equal.

$p + i\beta - \beta \neq p$

$\Rightarrow (i-1)\beta \neq 0$

$\Rightarrow i - 1 \neq 0$

$\Rightarrow i \neq 1$

Let, $i = 2$, then the string doesn't belong to the langu

Q2.

Q2. $s = 1^{p^3+p}$

$$\underbrace{1^{\alpha}}_{x}\ \underbrace{1^{\beta}}_{y}\ \underbrace{1^{p^3+p-\alpha-\beta}}_{z}$$

after pumping i times, we get

$1^{\alpha}\, 1^{i\beta}\, 1^{p^3+p-\alpha-\beta}$

$\Rightarrow 1^{p^3+p+i\beta-\beta}$; if this string is between two consecutive members of the language then it can not be in the language.

let $i = 2$,

$p^3+p+\beta < (p+1)^3 + (p+1)$

$\Rightarrow p^3+p+\beta < p^3+3p^2+3p+1+p+1$

$\Rightarrow \beta < 3p^2+3p+2$

as $\beta \leq p$, the above is always true.

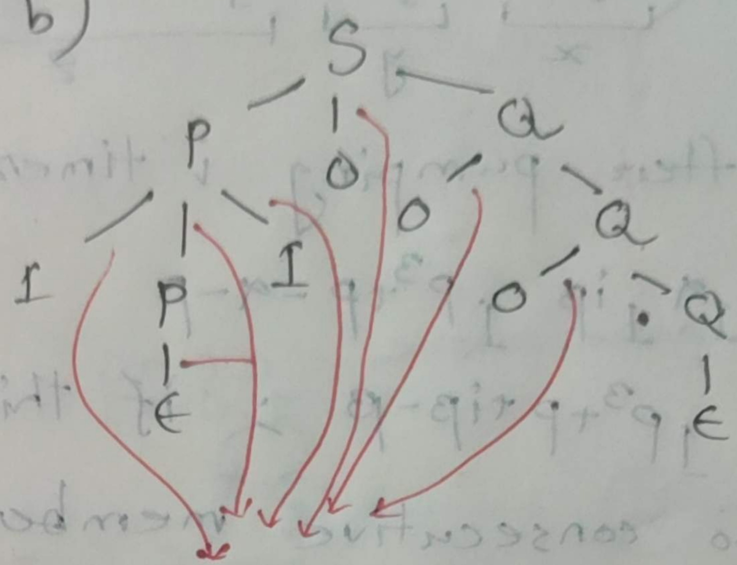
$\therefore 1^{p^3+p+\beta} \notin L$

Q3.

a)

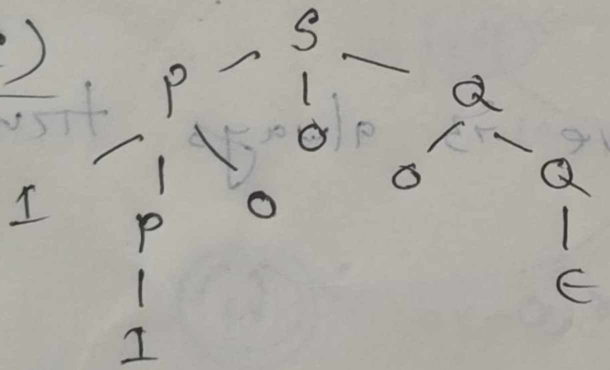
S
↓
POQ
↓
1P1OQ
↓
11OQ
↓
11OOQ
↓
11OOOQ
↓
11OOO

b)



You do not need to extend these edges!

c)



d) 11011

[the string has to contain exactly 1 zero]

Quiz 4

Q1. $w = x0y$ CFG

$$S \to 0S0 \mid 0S1 \mid 1S0 \mid 1S1 \mid 0$$

$w = 0^n 1^m$, $m = 2n+1$

$$S \to 0S11 \mid 1$$

Q3. a) F

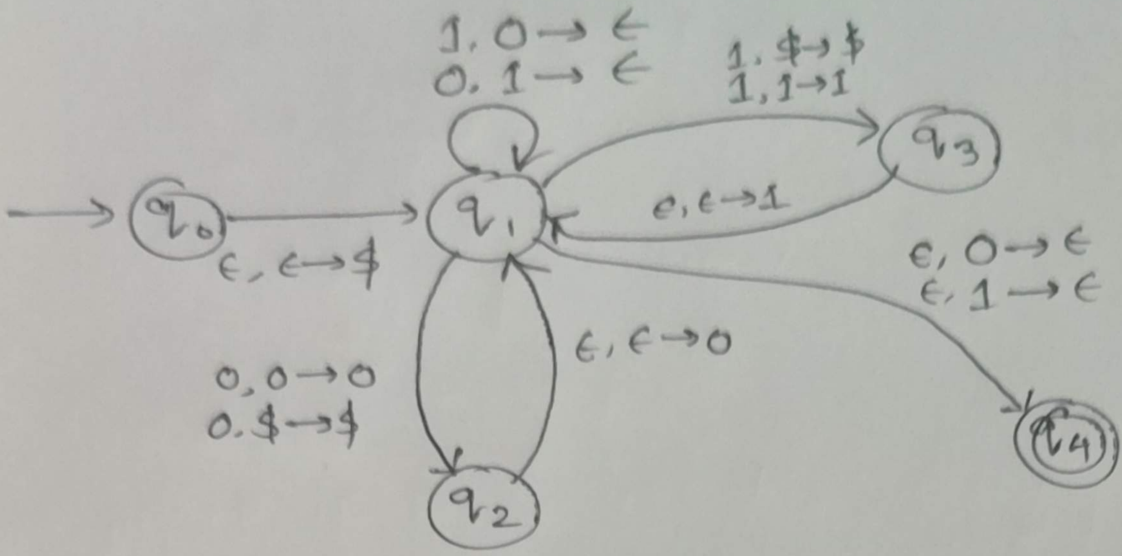
b) T

c) F

d) F

e) T

Q2. # 0's ≠ # 1's



Not a palindrome